AID843 Assignment 2 Report

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***Abstract*—As a part of the course AID843 - Spatio-temporal Data Analytics, this report contains the results of and challenges associated with the completion of tasks 1-3 of Assignment 2, covering temporal preprocessing, stationarity and autocorrelation analysis, and regression-based forecasting of the NYC Drinking Water Tank Inspections (2021-2024) dataset.**

I. Introduction

This report details the application of temporal preprocessing, statistical testing, and regression modeling techniques to develop a robust understanding of the time-varying behaviour present in the NYC Drinking Water Tank Inspections (2021-2024) dataset. The management of New York City's potable water is, naturally, an complex urban challenge. As established in Assignment 1, the city's rooftop drinking water tanks (DWTs) represent a vulnerability to necessary public infrastructure, and as a result, their inspection and audits are mandated by the NYC Department of Health and Mental Hygiene. While Assignment 1 examined the spatial distribution of inspection and violation patterns, this report examines how those patterns evolve over time.

To study temporal emergent properties, the following techniques have been employed:

- Temporal decomposition, detrending, and deseasonalization
- Stationarity testing via Augmented Dickey-Fuller and KPSS tests
- Autocorrelation and partial autocorrelation analysis
- Linear regression and ARIMA-family forecasting models

II. Preprocessing and Dataset adjustments

The dataset contains 16,562 records across 22 columns, each representing a single exception. Due to the pass-fail nature of a DWT inspection, two 'types' of records emerge: 1,412 records with a non-null Violation Code representing failed inspections, and the remaining 15,150 records representing passed ones. This split, while motivated by the spatial analysis in A1, has direct implications for temporal analysis.

Violation records carry a Date of Occurrence field at day-level precision, spanning November 24, 2021 to March 11, 2024 — a window of 29 months. This enables monthly aggregation into a proper time series, with the caveat that the distribution of inspections is highly uneven. Non-violation records, however, carry only an Activity Year field. At annual resolution, this produces just four data points (2021–2024), which is entirely insufficient for decomposition, autocorrelation analysis, or any regression model. The non-violation subset was therefore found to be unusable and is excluded from all subsequent analysis.

From the violation subset, three monthly time series were constructed: a violation count series (number of violations per month), a compliance lag series (average years elapsed between a building's last successful inspection and its recorded violation), and a borough-wise breakdown series. Missing months were filled via time-based interpolation to enforce continuous monthly frequency, a necessary precondition for statsmodels decomposition functions. This interpolation, however must be handled carefully as it can lead to significant 'hallucinated' autocorrelation, seasonality, or residuals.

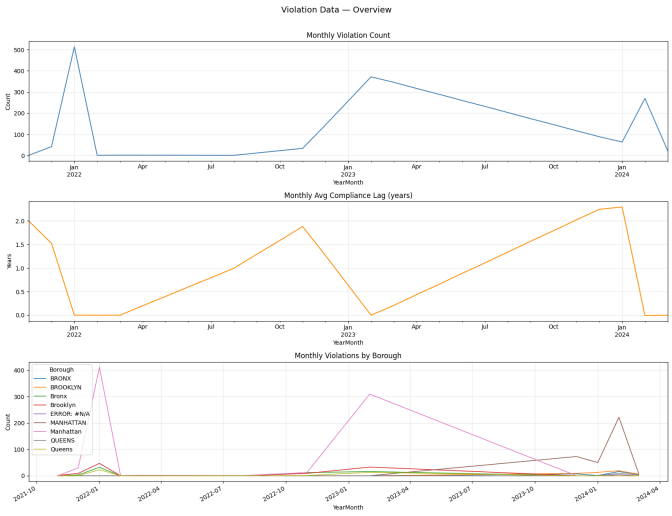

Fig. 1. The three time-series deemed interesting to study: violations per month, average compliance lag, and borough-wise violations. All plots are shown post-interpolation

The clear annual spikes in violation count indicate that either an incredibly high number of inspections are carried out between January and February, or that records are only updated once a year. The former is highly unlikely, seeing as how inspections require in-person activity, which becomes difficult in harsh winter conditions. The implications of this will be discussed in more details in the following sections.

III. Methods used for Analysis

A. Temporal Decomposition

Additive and multiplicative seasonal decomposition with period=12 was applied to the monthly violation count and compliance lag series. The trend component of the violation count series reveals a rise through 2022 into early 2023, followed by a decline, possibly the result of clearance of an inspection backlog caused by the COVID-19 pandemic.

The seasonal component, while mathematically extracted, must be interpreted with caution: given the Jan/Feb reporting concentration, the extracted seasonal signal seems to largely capture an administrative artefact rather than any genuine environmental or regulatory seasonality. The residual component displays large spikes precisely at Jan/Feb positions across years, confirming that the period-12 decomposition model does not adequately describe the discontinuity in the data.

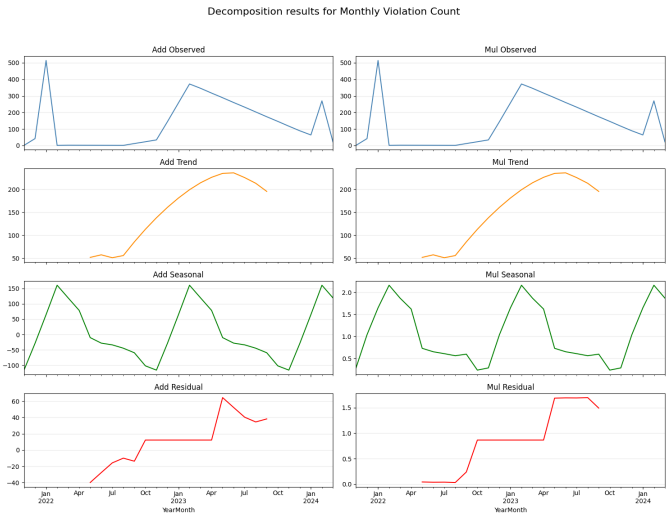

Fig. 2. Comparison between additive and multiplicative models for the violation count time series. Additive and multiplicative models appear nearly identical, as such the additive model was chosen to go ahead with.

The compliance lag series could only be decomposed additively, as it contains non-positive values in some months. Unlike the violation count data, its discontinuity does not seem to be the result of administrative issues, but rather reflective of a lack of the effective 'healthy' lifespan of a DWT, as the average compliance lag increases to 2 at the end of every year.

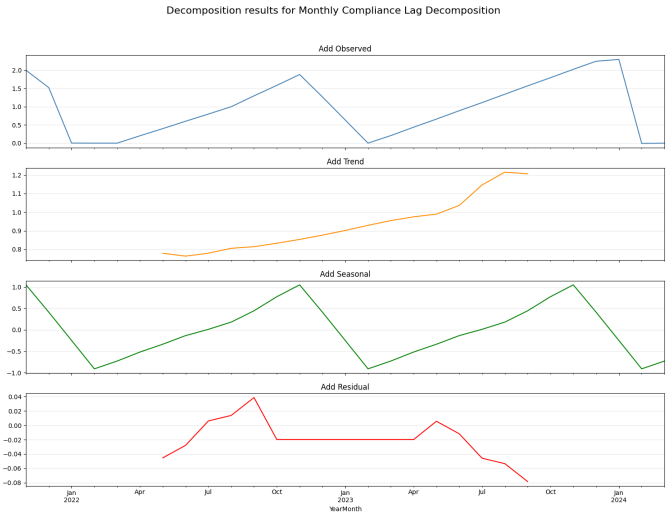

Fig. 3. Additive decomposition of the compliance lag time series, the multiplicative is impossible to use, as the compliance lag drops to 0 for several months in 2022 and 2024.

The passed inspections naturally only produced 4 timestamps, resulting in a resoltion that was much too low to be of any real analytic use. As a result, its decomposition yeilds incredibly misleading information at the month-wise resolution that the rest of the analysis methods rely on.

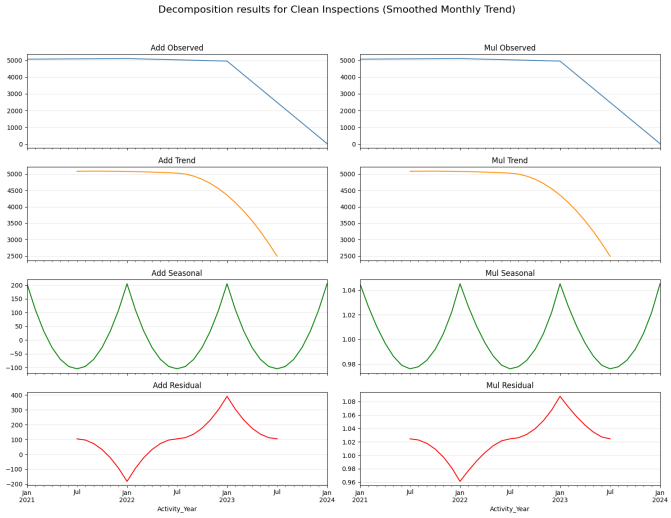

Fig. 4. Comparison between additive and multiplicative models for the violation count time series. Additive and multiplicative models appear nearly identical, as such the additive model was chosen to go ahead with.

B. *Stationarity Analysis*

Both the Augmented Dickey-Fuller (ADF) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests were applied to the original, detrended, deseasonalized, and first-differenced versions of the violation count series. ADF tests the null hypothesis of a unit root (non-stationarity), while KPSS tests the null of stationarity. Using both in tandem provides a more reliable verdict than either alone.

The results produced a coherent and analytically informative picture. The original series is stationary: ADF p = 0.034 and KPSS p = 0.1, with both tests in agreement. The detrended series is ambiguous, with ADF failing to reject non-stationarity (p = 0.177) while KPSS also fails to reject stationarity (p = 0.1), an artefact of statsmodels' function design. Most notably, the deseasonalized series is non-stationary: ADF p = 1.0 and KPSS p = 0.028. This result verifies the notion that purported seasonality seen during decomposition was indeed due to artefacts from the reporting process. Subtracting it removes the only periodic structure that was maintaining stationarity.

TABLE I
STATIONARITY ANALYSIS RESULTS FOR VIOLATION DATA

Series Type	ADF p-value	KPSS p-value	**Verdict**
Original	0.0344	0.1000	Stationary
Detrended	0.1771	0.1000	Ambiguous
Deseasonal	1.0000	0.0283	Non-Stationary
Differenced	0.0007	0.1000	Stationary

TABLE II
STATIONARITY ANALYSIS RESULTS FOR COMPLIANCE LAG DATA

Series Type	ADF p-value	KPSS p-value	Verdict
Original	0.0344	0.1000	Stationary
Detrended	0.1771	0.1000	Ambiguous
Deseasonal	1.0000	0.0283	Non-Stationary
Differenced	0.0007	0.1000	Stationary

C. Autocorrelation

ACF and PACF plots were generated for both the original and first-differenced series. The original ACF shows significant spikes at lags 1 and 12. Naively, these would suggest AR(1) structure and annual seasonality respectively. However, in light of the stationarity results and the visible reporting concentration, these spikes are more parsimoniously explained as artifacts of the Jan/Feb clustering — not evidence of genuine month-to-month or year-over-year temporal dependence.

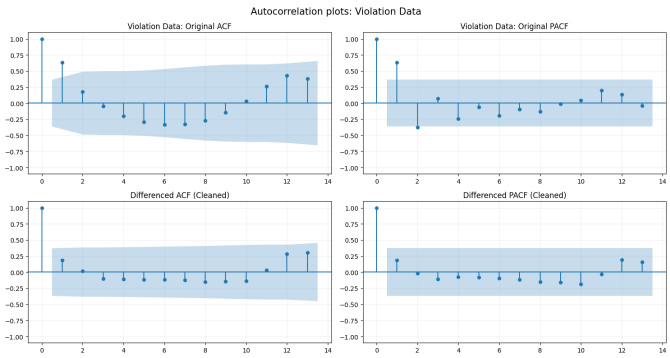

Fig. 5. ACFs and PACFs for the violation count series

Fig. 6. ACFs and PACFs for the compliance lag series

The PACF cuts off sharply after lag 3. After first-differencing, autocorrelation is largely eliminated, further confirming that the original series was already near-stationary and that the ACF spikes did not represent deep structural dependence.

D. Regression Models and Forecasting

An 80/20 chronological train-test split was applied, yielding a training period of November 2021 to September 2023 (23 months) and a test period of October 2023 to March 2024 (6 months). Normalization was fit exclusively on the training set to prevent data leakage.

Four models were implemented and evaluated:

- A linear trend model implemented as a single-layer PyTorch network mapping timestep index to violation count serves as the baseline.
- A linear regression model with lag features, also implemented in PyTorch, uses lag-1, lag-2, lag-3, and lag-12 values alongside a global trend index and cyclical month encodings (sin/cos) as inputs.
- An ARIMA model with order selected via AIC-minimizing stepwise search was fit without a seasonal component, as the reporting-artifact analysis established that the apparent seasonality is not genuine.
- A SARIMA model with m=12 was fit to test whether adding a seasonal component improves forecast accuracy.

In addition, normalisation was performed used min-max scaling fit on training data, with predictions inverse-transformed before evaluation. Models were evaluation on MAE and RMSE; MAPE was computed but found to be unreliable due to months with near-zero violation counts in the test set.

After training, it was seen that the non-seasonal ARIMA model fit best, with AIC for ARIMA(3,0,0) model being 25.044, compared to 25.260 and 27.076 for the non-seasonal and seasonal versions of SARIMA. In statistical modeling, the model with the lowest AIC is preferred as it provides the best balance between fit and simplicity.

In addition, the linear model with lag features showed an astonishingly low RMSE of 0.005217, indicating overfitting when compared to the significantly more reasonable 0.095438 of the linear trend model.

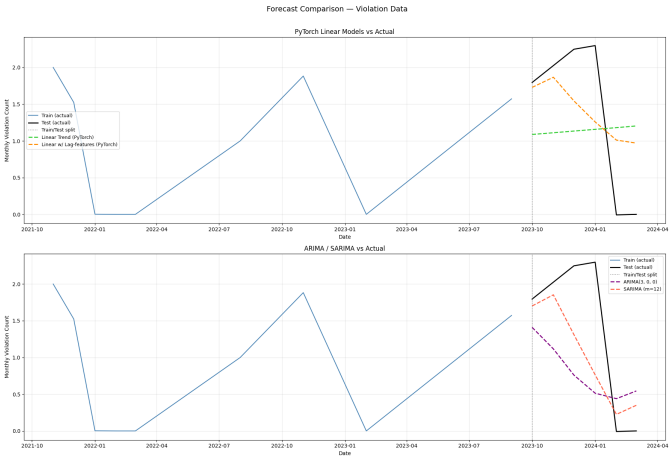

Fig. 7. Comparative forecasting of monthly water tank violations using PyTorch Linear models (Top) and ARIMA/SARIMA specifications (Bottom)

The top panel compares the PyTorch linear models against the actual inspection records. The Linear Trend serves as a simple baseline that captures a general upward trajectory but fails to account for the sharp cyclical variance inherent in the dataset. In contrast, the Linear Trend with Lag-features

demonstrates a significantly closer fit to the initial test-set spike, reaching a final training loss of 0.005217 compared to 0.095438 for the plain trend model. However, this extremely low loss indicates that the lag-feature model is likely overfitting.

The bottom panel displays the statistical baseline models. The SARIMA (m=12) model identifies the 12-month periodicity, producing a forecast that mimics the seasonal pulses found in the training data. Meanwhile, the ARIMA(3,0,0) was selected as the best non-seasonal fit with an AIC of 25.044, yet it predicts a sharper decay in violations. The stark cutoff observed in the actual data for early 2024 remains uncaptured by most models, confirming that predicting future compliance events from these administrative records is simply not present of easily learnable from the existing data.

IV. Discussion

A. *preprocessing,ipynb*

The notebook constructs the dataset split, parses Date of Occurrence into monthly periods, and builds three time series from the violation subset. A critical finding was that 91% of the dataset could not contribute to temporal analysis due to its annual resolution. The compliance lag series, derived as the difference between Activity Year and Compliance Year, emerged as a meaningful secondary series as it demonstrates the likely maximum lifetime of a DWT.

B. $autocorrelation_stationarity.ipynb$

The notebook implements ADF and KPSS testing across four transformations of the violation series, alongside ACF/PACF visualizations. These plots and analyses demonstrated that the apparent seasonality is simply a result of administrative artefacts, as deseasonalization removes stationarity. This is reinforced by the PACF's sudden cutoff at lag 3 and SARIMA's negligible improvement over ARIMA.

C. *regression.ipynb*

The notebook implements all four forecasting models, revealing a stark contrast in performance based on feature selection. A notable observation is the massive training loss gap between the two PyTorch models: the lag-feature model achieves a final loss of 0.005217, significantly lower than the 0.095438 recorded for the plain trend model. While this suggests that lag features carry a substantial within-sample predictive signal, the extremely low loss for the lag model indicates a high probability of overfitting. Given the small sample size and the artificial "shark-tooth" nature of the interpolated data, the model has likely memorized administrative reporting patterns rather than learning underlying environmental drivers.

V. Next Steps

A. *Count-Specific Models*

Monthly violation counts are non-negative integers. Linear regression and ARIMA treat the target as continuous and can in principle predict negative values, which are physically meaningless. Poisson regression or negative binomial models would be both statistically appropriate and more interpretable for this data type, and would handle the zero-count months that caused MAPE to fail entirely.

B. *External Regressors via SARIMAX*

The SARIMAX framework supports exogenous variables. Candidates include monthly precipitation data, temperature, and public budgets. Any of these could explain variance that pure autoregressive structure cannot.

VI. Declaration on the use of GenAI tools

I, Aditya S Rao, assert that GenAI tools have been used only for the following purposes: generating code for plots, learning library syntax and usage, and debugging.